DPPU Framework

Document 1: Origin

Da Vinci Dual-Center Geometry and the Derivation of phi = 4/pi

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phi = 4/pi = 1.273239544735163

1. The Starting Point

The Dynamic Pi-Phi Processing Unit (DPPU) did not begin with a machine learning problem. It began with a geometric observation about the human body. Da Vinci's Vitruvian Man encodes two geometric centers: a physical center at the navel, which anchors the circumscribed circle, and a structural center at the base of the pelvis, which anchors the inscribed square. These two centers are not the same point. The gap between them encodes a proportional constant that had not been formally extracted and applied to recurrent neural architecture.

The question this document answers: what is that constant, where does it come from geometrically, and does it do anything useful in a computational system?

2. Deriving phi = 4/pi from Vitruvian Geometry

The Vitruvian Man establishes a canonical proportional system. Normalize the figure so the circumscribed circle has radius r = 1. The inscribed square's side length, derived from Vitruvian proportions, is s = pi/2. The half-side of that square is pi/4. The ratio of the circle radius to the square half-side is the geometric constant of interest:

```
r = 1 (circle radius, normalized)
s = pi/2 (square side from Vitruvian proportions)
half-side = pi/4
phi = r / half-side = 1 / (pi/4) = 4/pi
phi = 4/pi = 1.273239544735163...
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This is not the golden ratio (1.618...) nor any standard mathematical constant. It is the ratio at which the Vitruvian circle and square are in proportion — the boundary between circular and rectilinear geometry embedded in the human figure. The two centers produce a natural scaling factor that describes the transition between oscillatory and linear dynamics.

3. The Dynamic Pi-Phi Field System

To confirm phi as a natural attractor rather than an imposed constant, a field simulation was constructed. Two dynamic functions interpolate between limiting states as a function of a tension parameter Delta, representing the degree of geometric folding or perceptual shift:

```
pi_dynamic(D) = 4 - (4 - pi_classical) * exp(-D)
phi_dynamic(D) = phi_classical * exp(-D) + (1 - exp(-D))
Omega(D) = (pi_dynamic(D) * phi_dynamic(D)) / (1 + D + eps)
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At D = 0: pi_dynamic approaches 4 and phi_dynamic approaches phi_classical = 4/pi. As D increases toward infinity: pi_dynamic approaches pi_classical (3.14159...) and phi_dynamic approaches 1. The product function Omega is maximized at a critical transition point D* that the simulation locates through gradient ascent.

Delta	pi_dynamic	phi_dynamic	Omega	Note
0.0	4.0000	1.2732	—	pure geometric state
0.5	3.429	1.210	peak region	approaching transition
0.816	3.284	1.184	maximum	critical point D*
2.0	3.178	1.097	declining	classical limit approach
5.0	3.143	1.013	near floor	near classical pi

Table 1. Field simulation values at selected Delta. Omega peaks at D ≈ 0.816, the natural transition between Vitruvian and classical geometry.*

The critical point D* ≈ 0.816 is where the field system is maximally active. At this point, the product phi_dynamic * pi_dynamic / (1 + D) reaches its peak, and the dynamic system transitions from the geometric state (pi = 4, phi = 4/pi) toward the classical state (pi = 3.14159, phi = 1). The DPPU cell operates at this transition point — using phi_classical = 4/pi as the fixed scalar that anchors the recurrent weight path at the geometric attractor.

4. From Geometry to Gradient Stabilization

The field simulation established phi = 4/pi as the natural attractor of the pi-phi product system. The question for machine learning is whether this constant has any utility as a scaling factor on recurrent weight paths. The hypothesis follows from the geometry:

phi = 4/pi is the ratio at which circular dynamics (spectral radius near 1, oscillatory) and linear dynamics (spectral radius near 0, contractive) are in proportion. Multiplying the recurrent weight path by phi biases the effective spectral radius of W_h toward this boundary — the operating point where gradients neither vanish exponentially nor explode.

This is the core architectural hypothesis. It is stated here as conjecture and tested systematically across Documents 3 through 9.

5. The DPPU Cell — First Statement

The cell definition that follows from this derivation:

```
h_t = tanh( W_x * x_t + phi * W_h * h_{t-1} + b )
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phi = 4/pi = 1.273239544735163 (fixed, not learned)
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phi is a fixed scalar. It is not initialized and then learned — it is permanently set to the geometric constant derived from the Vitruvian figure. It adds zero parameters. It costs nothing at inference. It is the only difference between DPPU and a standard Elman RNN.

6. What This Document Does Not Claim

Document 1 establishes the geometric origin of phi and states the architectural hypothesis. It does not provide experimental validation — that begins in Document 3. It does not prove that phi provides gradient stability — that is the subject of Documents 7 and 8. It does not claim phi is better than LSTM — that comparison is Document 4. The sole claim of this document is that phi = 4/pi is a natural geometric constant with a principled derivation, and that there is a coherent reason to test it as a recurrent scaling factor.

Core finding: phi = 4/pi emerges from the dual-center geometry of Da Vinci's Vitruvian Man as the ratio of circle radius to square half-side. It is confirmed as the natural attractor of the pi-phi field system at the critical transition D* ≈ 0.816. It is placed on the recurrent weight path of a standard Elman RNN with zero parameter cost. Experimental validation follows in Document 3.
